

DISSERTATION PROJECT

Predicting Population Performance and Genetic Variance to Guide Optimal Mate Allocation for Drought Tolerance in Maize

Vitor Bialetzki Cristiano

Advisor: Dr. Kaio Olimpio das Graças Dias

Viçosa - MG

Motivation

- **Importance of Maize:** Staple for 1 billion people; SDG 2 ("Zero Hunger").
- **Breeding Efficiency:** Prediction of mid-parental value (MPV) and genetic variance (V_g) can increase gain by 22% (Peixoto et al., 2025).
- **Goal:** Use *in-silico* simulations to optimize mate allocation under drought stress.
- **Question:** Can breeding simulations and population prediction accelerate the development of drought-tolerant maize hybrids while increasing selection efficiency and genetic gain?

Materials and Methods
Phenotypic and Genotypic Datasets

QGSL
Quantitative Genetics and Statistical Learning

Phenotypic Data

- **Hybrids:** 310 single-cross maize hybrids (Embrapa Milho e Sorgo).
- **Lines:** 190 inbred lines (17 Dent, 53 Flint, and 120 Group C).
- **Environments:** 8 environments total (2 years $\times$ 2 locations $\times$ 2 hydric regimes).
- **Locations:** Janaúba (MG) and Teresina (PI).
- **Conditions:** Well-watered (WW) and Water-stressed (WS).

Genotypic Data

- **Markers:** 47,127 SNP markers.
- **Quality Control:** MAF $>$ 5%; Missing genotypes $<$ 20%; Heterozygosity $<$ 5%.
- **Distribution:** Markers per chromosome ranging from 3,121 (Chr 10) to 7,705 (Chr 1).
- **Genetic Map:** Consensus map from MaizeGDB fitted via *smooth.spline* function.

QGSL
Quantitative Genetics and Statistical Learning

Statistical Model

$$y = 1\mu + X\beta + Zu + \epsilon \quad (1)$$

Where:

- $X_E\beta_E$: Environmental effects (year, location, water regime).
- $X_R\beta_R$: Replications within environments.
- $X_D\beta_D$ and $X_F\beta_F$: GCA (General Combining Ability) for the Dent and Flint groups.
- $X_H\beta_H$: SCA (Specific Combining Ability) for hybrid combinations.
- Zu: SNP marker matrix.
- Resíduos: $\epsilon \sim N(0, \text{Diag}(\sigma^2_\epsilon / w^2_i))$

Mate allocation and Population Prediction

QGSL
Quantitative Genetics and Statistical Learning

Metrics

- Population performance evaluated under well-watered (WW) and water-stressed (WS) conditions.
- Predicted population mean, GEBV, and Genetic variance.
- Heterotic Groups: *Flint* (53 + 120), *Dent* (17 + 120)
- RIL populations will be simulated using a modification of the sim.cross function from the R/PopVar package.

Optimal Population Value (OPV)

OPV represents the GEBV of the best possible progeny over unlimited generations:

$$OPV(S) = M \sum_{b=1}^{B} \max_{n \in S, m \in \{1,\ldots,M\}} \left(\sum_{l \in H(b)} A_{l,m,n} e_l \right) \tag{2}$$

- L: number of loci; M: ploidy level; B: haplotype blocks.
- A: binary allele matrix; e_l: effect of the major allele.
- Maximizes the capture of the best allelic combinations across blocks.

Optimal Haploid Value (OHV)

The OHV of an individual n is the GEBV of the best possible doubled haploid (DH):

$$OHV(n) = M \sum_{b=1}^{B} \max_{m \in \{1,\ldots,M\}} \left(\sum_{l \in H(b)} A_{l,m,n} e_l \right) \tag{3}$$

- Prioritizes genotypes with the highest potential to produce superior inbred lines.
- Focuses on the sum of the maximum GEBVs of the predefined blocks.

Thank you!



vitor.cristiano@ufv.br